

PYTHON PROGRAMMING INTERNSHIP

TASK 1 – PROJECT REPORT

Python Programming Fundamentals, Development Environment Setup & Problem-Solving Essentials

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Task: Task 1

Repository: python-programming-internship

1. Introduction

This project was completed as part of the Python Programming Internship Task 1. The objective of this task was to understand Python programming fundamentals, set up the development environment, practice problem-solving concepts, and build a simple command-line application.

During this task, I learned and practiced Python variables, data types, input and output, type conversion, operators, decision-making statements, loops, functions, modules, and debugging techniques. I also completed multiple programming exercises and developed a **Personal Expense Tracker** using Python.

The project was developed using Python and Visual Studio Code, with Git and GitHub used for version control and project management.

2. Development Environment Setup

The required development environment was prepared before starting the programming tasks.

The following tools were installed and configured:

- Python
- Visual Studio Code
- Python extension and Pylance in VS Code
- Git
- GitHub account and repository
- GitHub Desktop

Python and Git installations were verified through the terminal. A GitHub repository named **python-programming-internship** was created for the internship work.

The first Python program was also created and executed successfully to verify that the Python environment was working correctly.

3. Python Programming Fundamentals

The following Python fundamentals were studied and practiced during the task:

Variables and Data Types

I practiced creating variables and working with common Python data types such as strings, integers, floats, and booleans.

Input, Output and Type Conversion

I learned how to accept user input using `input()` and display results using `print()`. I also practiced converting input values using functions such as `int()` and `float()`.

Operators

I practiced:

- Arithmetic operators
- Assignment operators
- Comparison operators
- Logical operators
- Membership operators
- Identity operators

Comments and Naming Conventions

I practiced adding comments to Python programs and followed appropriate naming conventions to make the code easier to read and understand.

4. Decision Making and Loops

I practiced Python decision-making statements including:

- if
- if-else
- if-elif-else
- Nested if

These concepts were applied to different programming exercises.

I also learned and practiced loops:

- for loop
- while loop
- break
- continue
- pass

Loops were used to create multiplication tables, number patterns, repetition exercises, and other programs.

5. Functions

I learned how to create reusable functions in Python.

The following concepts were practiced:

- Defining functions
- Parameters
- Return values
- Variable scope
- Lambda functions

Functions helped me understand how larger programs can be divided into smaller and reusable sections of code.

6. Modules

I learned how to import and use Python modules.

The following modules were practiced:

- math
- random
- datetime

I used these modules for mathematical calculations, generating random values, and working with dates and time.

7. Debugging

I practiced identifying and fixing different types of errors in Python programs.

The following were covered:

- Syntax errors
- Runtime errors
- Logic errors
- Breakpoints
- Debug Console

I intentionally created errors to understand how they appear during program execution and then corrected them. I also used the VS Code debugger and Debug Console to inspect variables while the program was paused at a breakpoint.

8. Practice Exercises

As part of the programming practice, I completed several exercises based on the internship task, including:

- Personal Profile
- Simple Calculator
- Temperature Converter
- Simple Interest Calculator
- Multiplication Tables
- Prime Number Checker
- Password Strength Checker
- Menu-Driven Calculator
- Student Grade Calculator

- Electricity Bill
- ATM Menu Simulation
- Number Guessing Game

These exercises helped reinforce variables, input/output, conditions, loops, functions, operators, and problem-solving concepts.

9. Mini Project – Personal Expense Tracker

The main mini project developed during this task was a Personal Expense Tracker. The application is a command-line program that allows users to add, view, search, and analyze personal expenses.

The final version includes expense categories, date and time recording, category-based searching, highest and lowest expense identification, daily and monthly summaries, text-file saving, CSV export, and basic input validation. The program uses a list of dictionaries to store expense records and uses functions to organize the different operations. A while loop keeps the menu running until the user selects the Exit option. All major menu options were tested during development.

10. Project Documentation

The project was organized into separate folders and Python files for the different concepts and exercises.

Documentation prepared for the project includes:

- README.md
- Screenshots of programs and outputs
- Personal Expense Tracker source code
- Personal Expense Tracker flowchart
- Project report

The screenshots are organized inside the respective Day folders, and the Expense Tracker flowchart is stored in the project flowchart folder.

11. Learning Outcomes

Through this task, I gained practical knowledge of Python programming fundamentals and basic problem-solving.

I learned how to:

- Write and execute Python programs.
- Work with variables and different data types.
- Accept and process user input.
- Use operators and expressions.
- Apply decision-making statements.
- Use loops for repetitive tasks.
- Create reusable functions.
- Understand variable scope and lambda functions.
- Import and use Python modules.
- Identify and fix programming errors.
- Use the VS Code debugger.
- Organize a Python project.
- Use Git and GitHub for project management.
- Build a basic command-line application.

12. Conclusion

The Python Programming Internship Task 1 provided practical experience with Python programming fundamentals, development environment setup, debugging, and problem-solving.

By completing the exercises and developing the Personal Expense Tracker, I was able to apply the concepts learned throughout the task in practical programs. The project also helped me understand how individual Python concepts can be combined to create a functional command-line application.

Overall, this task strengthened my Python fundamentals and provided a foundation for developing more structured and reusable Python programs.